

財政學 HW4

1. (Rosen 16.1) According to estimates by Goolsbee and Petrin [2004], the elasticity of demand for basic cable service is -0.51 , and the elasticity of demand for direct broadcast satellites is -7.40 . Suppose that a community wants to raise a given amount of revenue by taxing cable service and the use of direct broadcast satellites. If the community's goal is to raise the money as efficiently as possible, what should be the ratio of the cable tax to the satellite tax? Discuss briefly the assumptions behind your calculation.

ANS

Assuming that all other commodities (except for cable and satellite television) were untaxed, then optimal tax policy suggests the commodities should be taxed according to the inverse elasticity rule. Goolsbee and Petrin (2004) find that the elasticity of demand for basic cable service is -0.51 , and the demand for direct broadcast satellites is -7.40 . Applying the inverse elasticity rule would imply that:

$$(t_{\text{BASIC}}/t_{\text{SATELLITE}})=(\eta_{\text{SATELLITE}}/\eta_{\text{BASIC}})=(7.40/0.51)=14.5$$

Thus, tax rates on basic cable should be 14.5 times higher than tax rates on satellite television because basic cable is inelastically demanded, while demand for satellite television is highly elastic. Among the assumptions that go into the inverse elasticity rule are that goods are neither complements nor substitutes, and that the elasticities are the Hicksian compensated elasticities rather than the Marshallian uncompensated elasticities. In this case, it is likely that the first of these assumptions is false – basic cable and satellite television are likely substitutes for each other. The Hicksian and Marshallian demand elasticities are likely to be close to each other because the income effects are likely to be small for this commodity.

2. (Rosen 16.3) “Peter the Great at one time levied a tax upon beards. He held that the beard was a superfluous and useless ornament. The tax is said to have been proportional according to the length of the beard and progressive according to the social position of its possessor” [Groves, 1946, p. 51]. Evaluate Peter's beard tax from the standpoint of horizontal equity.

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The beard tax was progressive because it was a function of social position, which was probably a good proxy for income. Because the tax attempted to treat people with the same income similarly, it satisfied some level of horizontal equity, even though it penalized those with a preference for facial hair. It's hard to know about the efficiency consequences unless one knows more about the price elasticity of demand for the privilege of having a beard. If the elasticity was small, then it would be an efficient tax.

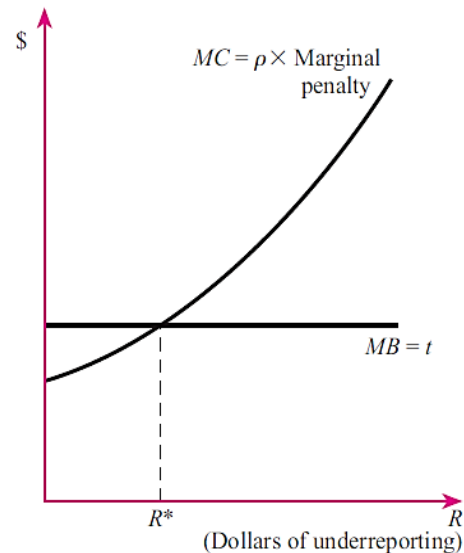
3. (Rosen 16.6) In 2008, New York State increased its tax on cigarettes by \$1.25 per pack. One commentator noted that “most cigarettes sold [in New York] will actually be trucked up from Virginia or shipped from China, by ‘butt-leggers’ who can make over \$1 million on each tractor-trailer load of smuggled smokes” [Fleenor, 2008]. Use Figure 16.5 to show how this behavior could have been predicted.

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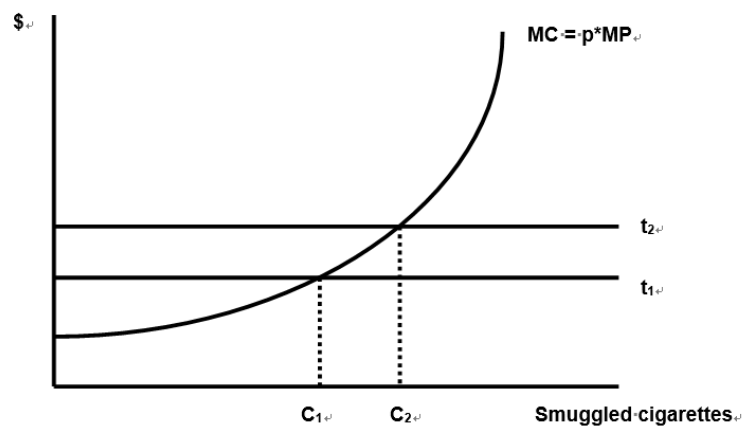
Figure 16.5

Tax evasion is positive

Given this tax rate, probability of being audited, and marginal penalty if caught cheating, the optimal amount of cheating is R^* , which is positive.



If the tax increases, the model implies that the number of smuggled cigarettes will increase because the marginal benefit of doing so (tax avoided) has increased.



4. (Rosen 16.8) Indicate whether each of the following statements is true, false, or uncertain, and explain why:

- A proportional tax on all commodities including leisure is equivalent to a lump sum tax.
- Efficiency is maximized when all commodities are taxed at the same rate.
- Average cost pricing for a natural monopoly allows the enterprise to break even, but the outcome is inefficient.
- Tom's workplace provides free access to a fitness room; Jerry's does not. Horizontal equity requires that Tom be taxed on the value of having access to the fitness room.

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(a) It is true that a proportional tax on all commodities (including leisure) is equivalent to a lump sum tax. To illustrate, consider the simplest example where there are only two goods: consumption goods and leisure. The budget constraint is equal to: $p_c C + wL = I$, where p_c and w are the prices of consumption goods and leisure, C and L are the quantities of consumption and leisure, and I is income. Then a proportional tax on all goods changes the budget constraint to: $(1-\tau)p_c C + (1-\tau)wL = I$, or rearranging, $p_c C + wL = I'$, where $I' = I/(1-\tau) < I$. Thus, a proportional tax on all goods does not change relative prices and is equivalent to taking away income. So it is equivalent to a lump sum tax.

(b) It is (usually) false that efficiency is maximized when all commodities are taxed at the same rate; this will not be true if leisure is untaxed. Imagine a more complicated budget constraint: $p_c C + p_f F + wL = I$. If leisure cannot be taxed, then a tax on commodities leads to a budget constraint of $(1-\tau)p_c C + (1-\tau)p_f F + wL = I$, which does change the relative price of leisure compared with food or consumption goods. Thus, it is not a lump sum tax. Instead, the inverse elasticity rule given in equation (16.9) would suggest that the ratio of the tax rates are inversely related to the ratio of the compensated demand elasticities for all commodities that can be taxed. That is, $(t_c/t_f) = (\eta_f/\eta_c)$.

(c) It is true that average cost pricing for a natural monopoly allows the enterprise to break even, but the outcome is inefficient. Figure 16.3 in the textbook shows the typical natural monopoly problem, with an initial fixed cost, and an ever-declining marginal cost curve. In this case, the average cost curve is always declining, but above the marginal cost curve. Setting $P=AC$ results in an output level of Z_A and zero economic profits. The figure illustrates, however, that the marginal benefit of more output exceeds the marginal cost, so the efficient level of production occurs at $P=MC$, or an output level $Z' > Z_A$. The deadweight loss is the area between the demand curve D_z and marginal cost curve, going from Z_A to Z' . If output were at the efficient level, however, there would be economic losses rather than zero profits.

(d) One notion of *horizontal equity* is that people in equal positions should be treated equally by the tax system. Under this traditional notion of horizontal equity, the fact that Tom's workplace provides free access to a fitness room suggests this kind of compensation should be taxed; Jerry pays "full taxes" on his compensation while Tom does not. Another notion of horizontal equity relies on the *utility definition of horizontal equity*. This concept says that if two individuals have the same utility without taxes, they should have the same utility with taxes, and the taxes should not affect the utility ordering. One implication of the *utility definition* is that any existing tax structure does not violate the notion of horizontal equity if individuals are free to choose their activities and expenditures. If Tom and Jerry have free choice between the two different jobs (and identical preferences), then the net after-tax rewards (including amenities) must be the same at both jobs; otherwise there would be migration. In this case, the before-tax wage on Tom's job

adjusts for the fact that there is a fringe benefit.

5. (Rosen 16.5) Suppose that a town is considering a project to install new underground water pipes. Some of the costs are fixed in the sense that they do not increase with an increase in the amount of water consumed. For example, pipes deteriorate over time, independent of the volume of water flowing through them. Hence, it is impossible to pay for the investment in the pipes by charging consumers an amount based on the marginal cost of the water they consume. Under these circumstances, what might an efficient pricing system look like?

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Efficiency generally requires setting price equal to MC, but doing so here would result in a price lower than the average cost, which would be unsustainable. In this case, the problem of paying for fixed costs could be solved by charging a price equal to MC and levying a lump sum tax to pay for the fixed costs associated with the pipes.

6. (Rosen 16.4) In 2008, the sales tax in Cook County, Illinois, was doubled. One journalist speculated that the tax increase was enacted because government officials sensed that the tax base was more captive than usual, since falling real estate prices and a tough job market made it difficult for citizens to move. Examine this situation through the lens of optimal tax theory. Is the increase in the sales tax efficient? Is it fair?

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Given the reduced ability to avoid the tax because of the captive tax base, the increased sales tax appears to be tax efficient, although the tax may result in lower purchases of non-necessities. The tax might, however, be unfair because individuals with lower incomes because they tend to spend more of their income resulting in higher taxation.